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Fault Diagnosis of PMSM Motors using Convolutional Neural Networks on Wavelet Scalogram Images

Aleppo University — Faculty of Electrical & Electronic Engineering Department of Mechatronics Engineering

Research seminar report

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Companion chapter — read first. The electrical-machines and control -engineering background (PMSM construction & operation, the FOC drive, each fault with its causes, traditional vs automated detection, the proposed method's improvements, and a PMSM-vs-PMDC/BLDC/induction comparison) is in **engineering-**

background.pdf with supporting diagrams. This report focuses on the signal-processing and CNN pipeline.

Abstract

Permanent Magnet Synchronous Motors (PMSMs) are the workhorses of modern electric drives — electric vehicles, robotics, industrial servo systems and aerospace actuators — because of their high power density, high efficiency and excellent controllability. Like all electrical machines they are subject to faults, the most common and most insidious of which is the **inter-turn stator winding short circuit**: a small number of shorted turns that, if undetected, escalates into a catastrophic winding failure within minutes. Early, automatic detection of such faults is therefore of large practical value.

This report presents a complete, reproducible pipeline for **PMSM fault diagnosis** that (i) collects motor current and vibration signals, (ii) converts short signal windows into **Continuous Wavelet Transform (CWT) scalogram images**, and (iii) classifies those images with a **Convolutional Neural Network (CNN)**. The approach turns a one-dimensional time-series problem into a two-dimensional image-recognition problem, allowing the CNN to learn fault signatures automatically rather than relying on hand-engineered features.

The pipeline is implemented end-to-end in Python (with an alternative MATLAB path), is covered by 38 unit tests and continuous integration, and is trained and evaluated on the **real KAIST PMSM stator-fault dataset**. On held-out recordings, **vibration scalograms separate healthy from inter-turn-faulty motors with a balanced accuracy of 1.00**, while **current scalograms achieve only 0.69** — a clear, physically-sensible result showing that vibration carries a much stronger inter-turn signature than stator current at the available fault severities. We report all results with balanced accuracy and macro-F1 (rather than raw accuracy) because the test set is class-imbalanced, and we are explicit about the principal limitation of the public data: only four distinct healthy

recordings exist, so generalisation of the perfect vibration score cannot yet be fully guaranteed.

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1. Introduction

1.1 Motivation

Condition monitoring and fault diagnosis of electric machines is a cornerstone of predictive maintenance. A drive that can recognise its own deterioration can warn an operator, de-rate itself, or trigger a controlled shutdown before a minor fault becomes a destructive one. For PMSMs the economic and safety stakes are high: they are used in traction (electric vehicles), in surgical and industrial robots, in aircraft actuators and in wind-turbine pitch systems, where unplanned failure is expensive or dangerous.

Traditional fault-diagnosis methods rely on **expert-designed features** — for example, tracking specific harmonic amplitudes in the stator current spectrum (Motor Current Signature Analysis, MCSA). These methods are powerful but brittle: they require a human to know in advance *which* frequency to watch, and they degrade when operating conditions (speed, load) vary. Modern **deep learning** offers an alternative: a CNN can learn the relevant features directly from the data, provided the data is presented in a form the CNN can exploit. The central idea of this project is to present motor signals as **time-frequency images** (scalograms), which expose fault signatures as visual patterns that a CNN is ideally suited to recognise.

1.2 Objective

The main task assigned for this seminar is:

Design a CNN-based model that analyses Wavelet Scalogram images produced from PMSM signals, in order to classify operating states and detect faults.

Concretely, the project must: study the relevant theory; obtain PMSM signals (current and/or vibration) under different operating states; apply the CWT to generate scalogram images organised by class; build and train a CNN that classifies these images; and finally test, analyse and improve the model. The deliverables are a labelled image dataset, training code, a final CNN model, result figures, this report and an accompanying presentation.

1.3 Contributions

Beyond the minimum requirements, this work delivers:

- A **fully reproducible, MATLAB-free Python pipeline** driven by a single configuration file and a manifest that links every signal segment to its scalogram, label and data split.

- Use of a **real, high-sample-rate public dataset** (KAIST) with **two signal channels** (current *and* vibration), plus a synthetic generator and MATLAB simulation scripts as alternative signal sources.
 - A **leakage-free, recording-grouped** train/validation/test split, and a principled treatment of **class imbalance** (the most important practical issue with the real data).
 - A **dual-branch fusion CNN** that combines the current and vibration channels.
 - **Software-engineering rigor** unusual for a student project: 38 unit tests, continuous integration, and an honest, metric-appropriate reporting of results.
-

2. PMSM Motors and their Faults

2.1 Operating principle

A Permanent Magnet Synchronous Motor is a three-phase AC machine in which the rotor carries permanent magnets and the stator carries a three-phase winding. When the stator windings are energised with a balanced three-phase current, they create a rotating magnetic field; the rotor magnets lock onto this field and rotate **synchronously** with it. Unlike an induction motor, there is no slip and no rotor current, which is what gives the PMSM its high efficiency and high torque density.

The fundamental energy balance of the machine equates electrical and mechanical power, $v \cdot i \approx T \cdot \omega$: voltage is associated with speed (ω) and current with torque (T). This is why current is the natural quantity to control for torque, and why current waveforms carry rich information about the electromechanical state of the machine.

PMSMs are almost always driven by a **Field-Oriented Control (FOC)** scheme.

FOC uses the rotor position (from an encoder or a sensorless estimator) to transform the three phase currents into two orthogonal components: a **direct-axis** current i_d (aligned with the rotor flux, ideally held near zero) and a **quadrature-axis** current i_q (orthogonal to the flux, which produces torque). Two PI controllers regulate i_d and i_q , and the resulting voltage demands are synthesised by a PWM inverter (three half-bridges, with a current sensor on each phase). FOC is important to this project for two reasons: it is the context in which real PMSM current signals are produced, and it shapes those signals (the controller actively attenuates some fault signatures, which is one reason faults can be easier to see in vibration than in current).

2.2 BLDC vs PMSM

The PMSM is closely related to the Brushless DC (BLDC) motor. The practical distinction is the shape of the back-EMF and the drive current: a BLDC motor has a **trapezoidal** back-EMF and is driven with quasi-square (trapezoidal) currents, whereas a PMSM has a **sinusoidal** back-EMF and is driven with **sinusoidal** currents (via sinusoidal PWM). The sinusoidal nature of PMSM currents makes frequency-domain and time-frequency analysis particularly meaningful.

2.3 Common fault types

Electrical-machine faults are usually grouped as electrical, mechanical and magnetic:

- **Inter-turn stator short circuit (ITSC).** A breakdown of the insulation between adjacent turns of the same phase winding creates a shorted loop. A large fault current circulates in the shorted turns, producing local heating that accelerates further insulation failure. ITSC is the most common stator fault and the primary fault studied in this project. It introduces characteristic asymmetries and additional harmonic/side-band components in the current and a distinct vibration signature.

- **Demagnetization.** Partial loss of rotor magnet strength (from overheating, ageing, or large demagnetising currents) reduces the back-EMF and torque constant and introduces sub-harmonic and side-band components around the fundamental. In this project demagnetization is represented only by the synthetic generator, because it is absent from the public real dataset.
- **Overload / thermal stress.** Sustained operation above rated load raises the fundamental current and changes the harmonic content; it is a degraded *operating state* rather than a hard fault, and is likewise represented synthetically here.
- **Mechanical faults** (bearing wear, eccentricity, misalignment) are outside the scope of this project but are naturally addressed by the same scalogram+CNN methodology, especially on the vibration channel.

2.4 Why current and vibration

Two complementary sensing modalities are used:

- **Stator current** is essentially free to measure — current sensors already exist in every FOC drive — which makes current-based diagnosis very attractive industrially. However, the closed-loop controller partially suppresses fault signatures, and inter-turn signatures in the current can be weak at low severities.
- **Vibration** requires an accelerometer but responds directly to the electromagnetic and mechanical asymmetries a fault produces, often giving a much stronger and cleaner fault signature. Our results bear this out.

3. Signal Processing Background

A motor signal $x(t)$ is a one-dimensional time series sampled at a rate f_s . The simplest description is in the **time domain** (amplitude versus time), which shows *when* things happen but not *what frequencies* are present. The classical tool for the latter is the **Fourier Transform**, which decomposes a signal into a sum of sine waves and reports the amplitude of each frequency.

The Fourier Transform, however, is **blind to time**: it tells us which frequencies are present over the whole record, but not when they occur. For a *stationary* signal (statistics constant in time) this is fine. Motor fault signals are frequently **non-stationary** — transient, load-dependent, or modulated — and for these the Fourier Transform is inadequate.

This limitation is fundamental, not merely practical. The **uncertainty principle** of signal processing states that the product of time resolution and frequency resolution is bounded below: $\Delta t \cdot \Delta f \geq \text{constant}$. We cannot know *both* the exact time and the exact frequency of an event with arbitrary precision. Any time-frequency method must therefore make a trade-off, and different methods make it in different ways. The Short-Time Fourier Transform (STFT) fixes a single window length and therefore a single, frequency-independent trade-off. The **Wavelet Transform** makes a smarter, frequency-dependent trade-off, and is the method used in this project.

4. The Wavelet Transform and Scalograms

4.1 From sines to wavelets

The Fourier Transform correlates the signal against infinitely-long sine waves. A **wavelet** (“little wave”) is instead a short-lived oscillation that is **localised in time**. Replacing the eternal sine with a localised wavelet as the analysis

function is the key idea of the wavelet transform: because the analysis function itself has a finite extent in time, the result can tell us *where* in time each frequency component occurs.

A function $\psi(t)$ qualifies as a (mother) wavelet if it satisfies two conditions:

1. **Zero mean (admissibility):** its integral over all time is zero ($\int \psi(t) dt = 0$). Equivalently, the wavelet has no zero-frequency / DC component.
2. **Finite energy:** $\int |\psi(t)|^2 dt < \infty$, i.e. the wavelet is localised and dies away, unlike the sine which persists forever.

Many wavelet families exist — Haar, Daubechies, Coiflet, Symlet, Meyer, Mexican hat, Shannon, Gaussian and **Morlet**. The Morlet wavelet is the standard choice for time-frequency analysis of oscillatory signals and is used here. In simple terms the (real) Morlet wavelet is a cosine multiplied by a Gaussian bell, $\psi(t) \approx k \cdot \cos(\omega_0 t) \cdot e^{(-t^2/2)}$ — an oscillation that is “damped” so that it is concentrated around $t = 0$. In practice a **complex** Morlet is used, $\psi(t) = k \cdot e^{(i\omega_0 t)} \cdot e^{(-t^2/2)}$, whose magnitude gives a smooth measure of the local energy at a given frequency (avoiding the zero-crossings that a purely real wavelet would produce). In this project we use the PyWavelets complex Morlet `cmor1.5-1.0` (bandwidth 1.5, centre frequency 1.0); the MATLAB path uses the analytic Morlet `amor`.

4.2 The Continuous Wavelet Transform

The Continuous Wavelet Transform (CWT) generates a family of “daughter” wavelets from the mother wavelet by two operations:

- **Translation** (a time knob b): the wavelet slides along the time axis, $\psi(t - b)$, selecting *where* we look.
- **Scaling** (a frequency knob a): the wavelet is stretched or compressed, $\psi(t/a)$. A **compressed** wavelet (small a) oscillates rapidly and matches **high** frequencies; a **stretched** wavelet (large a) matches **low** frequen-

cies.

Combining both gives the daughter wavelet $\psi_{\{a,b\}}(t) = \psi((t - b)/a)$. The CWT coefficient $T(a, b)$ is obtained by correlating the signal with this daughter wavelet — mathematically an integral of the product of the signal and the (conjugated) wavelet. This is exactly a **dot product**, which measures **similarity**: $T(a, b)$ quantifies how strongly the signal resembles a wave of the given scale (frequency) at the given time. Sweeping b for a fixed a is a **convolution**; sweeping all (a, b) produces a full two-dimensional map. Where the Fourier Transform maps one dimension (time) to one dimension (frequency), the wavelet transform maps one dimension to a **two-dimensional** time-frequency representation.

4.3 The scalogram

The **scalogram** is the visualisation of $|T(a, b)|$ — the magnitude of the CWT coefficients — as a colour image, with time on the horizontal axis, scale (frequency) on the vertical axis, and colour encoding the **energy** of each frequency at each instant. Bright regions indicate strong, localised oscillatory energy; their position and shape encode the signal’s time-frequency structure, which is precisely what carries fault information.

The scalogram visibly reflects the uncertainty principle: at **low** frequencies the analysis has good frequency resolution but poor time resolution (broad, horizontally-smeared features), while at **high** frequencies it has good time resolution but poor frequency resolution (sharp in time, blurred in frequency). These are the “Heisenberg boxes” of the representation. Crucially, this frequency-dependent resolution is well matched to physical signals, where slow phenomena are usually persistent and fast phenomena are usually transient.

4.4 Why scalograms for CNNs

By turning each signal window into a scalogram **image**, we convert the fault diagnosis problem into an **image-classification** problem. This is powerful because (i) fault signatures become spatial patterns (bands, side-bands, texture) that survive moderate changes in operating point, and (ii) it lets us bring the full machinery of modern computer vision — convolutional networks — to bear on a signal-processing task.

Figure 4.1 shows real scalograms from the KAIST dataset for both channels and both classes. Note how the inter-turn fault enriches the time-frequency texture, especially in the vibration channel.

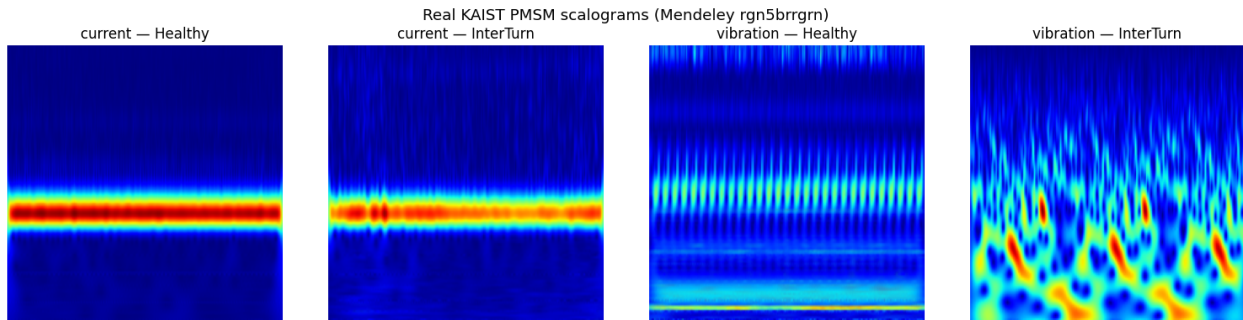


Figure 1: Figure 4.1 — Real KAIST scalograms: current and vibration, healthy vs inter-turn.

Figure 4.1 — Real KAIST scalograms (current and vibration, healthy vs inter-turn). The current channel is dominated by the fundamental band; vibration shows much richer structure that the CNN can exploit.

5. Convolutional Neural Networks

5.1 What a CNN is

A Convolutional Neural Network is a deep-learning architecture specialised for grid-structured data such as images. To a computer an image is a grid of num-

bers (pixel intensities); a CNN processes that grid through a sequence of layers that automatically learn to detect spatial patterns — from simple edges and textures in early layers to complex, class-specific structures in later layers. CNNs are loosely inspired by the organisation of the biological visual cortex.

5.2 The layers

A typical CNN passes the input image through the following stages:

1. **Convolutional layer.** Small learnable matrices (“filters” or “kernels”) slide across the image; at each position the filter is multiplied with the underlying pixels and summed, producing a **feature map** that highlights where a particular pattern (an edge, a band, a texture) occurs. Many filters per layer produce many feature maps.
2. **Activation (ReLU).** The Rectified Linear Unit sets negative values to zero, introducing the non-linearity that lets the network model complex patterns.
3. **Pooling layer.** Max-pooling downsamples each feature map by keeping only the maximum value in each small region, reducing spatial size and computation while retaining the strongest responses and adding a degree of translation invariance.
4. **Flattening / global pooling.** After several conv-pool blocks, the 2-D feature maps are reduced to a 1-D vector — either by flattening (concatenating all values) or by **global average pooling** (averaging each feature map to a single number, which is lighter and more robust to overfitting).
5. **Fully-connected (dense) layer.** A conventional neural-network layer performs the final reasoning on the feature vector and outputs class scores, normalised by a **softmax** into class probabilities.

5.3 Why CNNs (and their costs)

CNNs solve two problems that defeated earlier fully-connected networks on images. First, **parameter sharing** — the same filter is reused across the whole image — keeps the parameter count manageable, where a fully-connected network would need a separate weight for every pixel-neuron pair. Second, convolution **preserves spatial relationships** between neighbouring pixels, which flattening into a single row would destroy. The chief costs are that CNNs are **data- and compute-hungry**, can fail to generalise to transformations they were not trained or augmented for, and are relatively opaque (“black box”). In this project we mitigate the data cost with data augmentation, class balancing and a deliberately small architecture, and we are explicit about the generalisation limits imposed by the dataset.

6. Data Preparation

Data preparation is the heart of this project and the stage with the most design decisions. Every parameter mentioned below lives in a single `config.yaml`, and every produced segment is recorded in a single `data/manifest.csv` that links `signal` → `scalogram` → `label` → `split`. This manifest is the single source of truth for all downstream stages.

6.1 Signal sources

Three interchangeable signal sources feed the same pipeline:

Source	Channels	Tooling	Role
KAIST real dataset	current + vibration	Python loader	primary, reported results

Source	Channels	Tooling	Role
Synthetic generator	current	Python (simulate.py)	software validation; the only source of Demagnetization/Overload
MATLAB simulation	current	FOC + Simscape scripts	optional physics-based signals

Primary real dataset — KAIST. We use the *Vibration and Current Dataset of Three-Phase PMSM with Stator Faults* (Mendeley rgn5brrgrn, DOI 10.17632/rgn5brrgrn.5, CC-BY-4.0). It contains stator **current sampled at 100 kHz** and **vibration at 25.6 kHz**, recorded on 1.0 / 1.5 / 3.0 kW motors, under **normal**, **inter-turn** and **inter-coil** conditions across several fault severities, stored as TDMS files. The high sample rates make it well suited to time-frequency analysis. Inter-turn and inter-coil are both stator-winding short faults and are mapped to a single InterTurn class; 0 %-severity recordings are the Healthy class.

Secondary dataset — Zenodo (reference only). The *Comprehensive Dataset for Fault Detection and Diagnosis in Inverter-Driven PMSM Systems* (Zenodo 13974503) was audited but **not used for scalograms**: it is tabular sensor data sampled at only 10 Hz (Nyquist 5 Hz), which carries no useful time-frequency content for the CWT. It is retained as a reference for the discussion of data quality versus method fit.

Synthetic generator. A reproducible generator (python/simulate.py) builds current signals for all four classes with physically-motivated MCSA signatures (fundamental and odd harmonics for healthy; elevated 3rd/5th harmonics and

a side-band for inter-turn; sub-harmonic and $f_0 \pm k \cdot f_r$ side-bands for demagnetization; raised fundamental, even harmonics and noise for overload). It provides instant end-to-end validation and the only examples of the Demagnetization and Overload classes.

6.2 Classes

Four classes are defined in the configuration: **Healthy, InterTurn, Demagnetization, Overload**. The real KAIST data covers the first two; Demagnetization and Overload appear only in the synthetic data. The reported real-data results are therefore a **two-class** Healthy-vs-InterTurn problem, with the four-class configuration retained for the synthetic experiments and for forward compatibility.

6.3 Ingestion and segmentation

Each raw recording is read, then **decimated** to a common target rate of **10 kHz** (an anti-aliased FIR decimation from 100 kHz for current and from 25.6 kHz for vibration). Decimation makes scalogram sizes tractable while retaining all frequency content relevant to the fault (up to 5 kHz). Each recording is then cut into **0.5-second windows with 50 % overlap**; each window becomes one classified sample. Because the raw recordings are long (hundreds of seconds), a single recording would yield well over a thousand windows; to keep the dataset balanced in *recording* terms and the scalogram count tractable, at most **50 segments per recording** are kept, sampled at an even stride so they span the whole recording rather than just its opening seconds.

6.4 Scalogram generation

Every segment is converted to a scalogram with PyWavelets: the complex Morlet `cmor1.5-1.0` is applied over **128 scales** spaced geometrically between 4 and 256, the magnitude is normalised, mapped through the `jet` colourmap, and saved as a **224 × 224 RGB PNG** under `data/scalograms/<channel>/<class>/`.

This class-labelled folder layout satisfies the requirement to “save the images in folders classified by state” and is exactly the layout image classifiers expect. The whole real dataset produced **3,150 scalograms** (1,550 current + 1,600 vibration). An equivalent MATLAB path (matlab/scalogram/) produces the same layout using the Wavelet Toolbox.

6.5 Leakage-free train / validation / test split

A naïve random split of *segments* would place windows from the same recording in both the training and test sets. Because adjacent windows of one recording are highly correlated, this **data leakage** would inflate the test score and give a falsely optimistic result. We therefore split by **recording**: all segments of a recording are assigned to the same split (70 % train / 15 % validation / 15 % test), stratified by class, and the split is done **independently per channel** so that each channel is properly represented. A subtle but important refinement was required: with only four healthy recordings, ordinary rounding sent all of them to the training set, leaving validation and test with no healthy examples. The split routine was therefore modified to **guarantee at least one recording per split for any class that has enough recordings**, so that every class is evaluable. This behaviour is covered by unit tests.

6.6 Class imbalance and balancing

The real dataset is severely imbalanced: only **4 distinct healthy recordings** exist against roughly **60 fault recordings**. Trained naïvely on this distribution, the CNN learns to predict “InterTurn” for everything and achieves a deceptively high 80 % raw accuracy while **never detecting a healthy motor** (healthy recall 0.00). To prevent this collapse we **undersample the majority class in the training and validation splits only**, so the model receives a balanced learning signal; the **test split is deliberately left at its natural distribution** so that reported metrics reflect reality. Because raw accuracy

is misleading on an imbalanced test set, we report **balanced accuracy** (the mean of per-class recalls) and **2-class macro-F1** as the headline metrics. Data augmentation (random horizontal flips) is applied to the training images, and inverse-frequency class weighting is also available.

7. CNN Model and Training

7.1 Baseline architecture

The single-channel classifier is a compact CNN:

Input 224×224×3

→ Conv2D(32, 3×3) → ReLU → MaxPool2×2
→ Conv2D(64, 3×3) → ReLU → MaxPool2×2
→ Conv2D(128, 3×3) → ReLU → MaxPool2×2
→ Flatten → Dropout(0.5)
→ Dense(128) → ReLU
→ Dense(num_classes) → Softmax

Three convolution-pooling blocks (32, 64, 128 filters) extract a hierarchy of features; a 50 % **dropout** before the dense layer combats overfitting; a softmax output produces class probabilities. The architecture is intentionally small, appropriate to a dataset of a few thousand images.

7.2 Fusion architecture

To exploit both channels simultaneously, a **dual-branch fusion** model (`build_fusion_cnn`) runs two independent convolutional branches — one for the current scalogram, one for the vibration scalogram — each ending in **global average pooling**, then concatenates the two feature vectors and classifies the result through a shared dense head. Global average pooling

(rather than flattening) is used in the fusion branches because flattening the $26 \times 26 \times 128$ feature maps of two branches produces an enormous dense layer that exhausts memory; global average pooling is the standard, far lighter choice and adds regularisation.

7.3 Training setup

All models are trained with the **Adam** optimiser and **sparse categorical cross-entropy** loss. **Early stopping** (patience 5, restoring the best weights) prevents overfitting and avoids wasting computation. Training uses a batch size of 32 (16 for the heavier fusion model) for up to 30 epochs. Reproducibility is ensured by a fixed random seed (42) throughout ingestion, splitting, balancing and shuffling. TensorFlow 2.16 / Keras is used; the entire run executes on CPU.

7.4 Pairing for fusion

The fusion model needs paired current and vibration scalograms. The KAIST dataset stores the two modalities as separate recordings of the same operating condition; since both are decimated to the same rate and segmented identically, segment k of the current recording and segment k of the vibration recording cover the same fraction of the run — an approximate temporal pairing. To avoid leakage in the fused problem, the train/val/test split for fusion is assigned at the **condition level**, so that both modalities of a condition always fall in the same split.

8. Results and Analysis

All results below are on the **real KAIST data, held-out recordings**, the Healthy-vs-InterTurn problem, with balanced training and a natural test set.

8.1 Single-channel and fusion results

	Test	Balanced		Healthy	InterTurn
Channel	accuracy	accuracy	Macro-F1	recall	recall
Vibration	1.00	1.00	1.00	1.00	1.00
Fusion (current+vibration)	0.80	0.88	0.76	1.00	0.75
Current	0.50	0.69	0.49	1.00	0.37

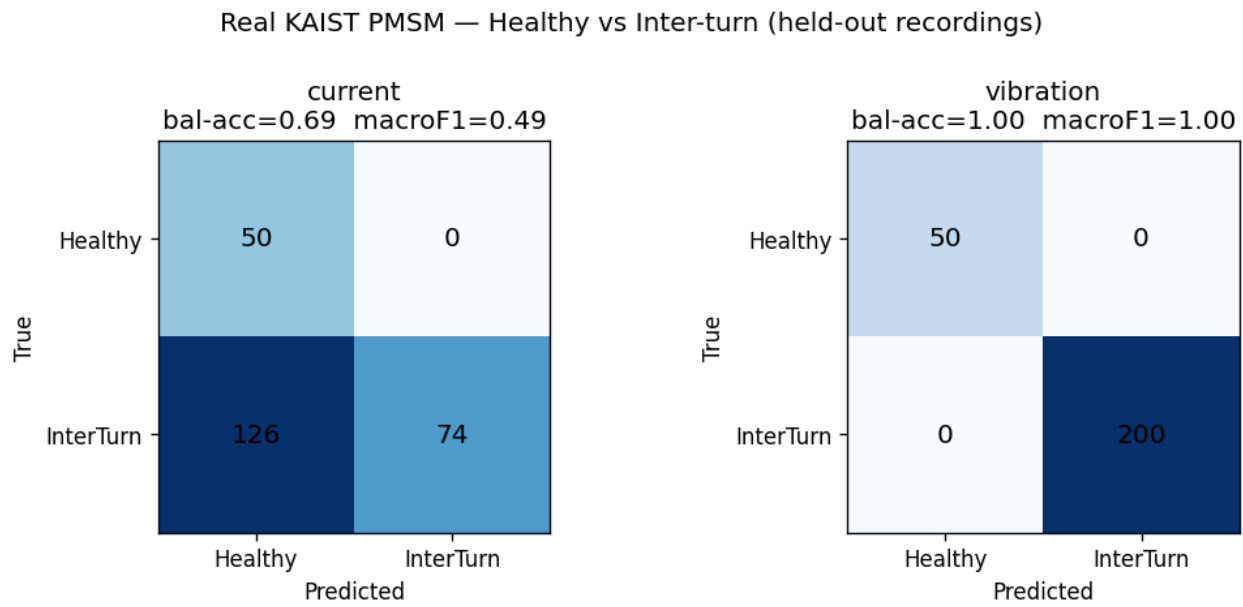


Figure 2: Figure 8.1 — Confusion matrices for the current and vibration channels.

Figure 8.1 — Test-set confusion matrices (held-out recordings). Left: current — all faults misread or split. Right: vibration — perfect separation. Each title shows balanced accuracy and macro-F1.

Example real scalograms are in Figure 4.1; machine-readable metrics in `results/real_metrics.json` and `results/report_{current,vibration,fusion}.json`.

8.2 Discussion

Vibration is the strong channel. The vibration CNN classifies every held-out test segment correctly (balanced accuracy 1.00). This is consistent with the physics: an inter-turn short produces a pronounced electromechanical asymmetry that manifests strongly in vibration, and the vibration channel is not subject to the suppressing action of the FOC current controllers. The example scalograms make this visible — the inter-turn vibration scalogram is markedly richer in time-frequency texture than the healthy one.

Current is the weak channel. The current CNN reaches only 0.69 balanced accuracy. After balancing it detects every healthy case (healthy recall 1.00) but misses about 63 % of faults (InterTurn recall 0.37). In other words, the inter-turn signature in the stator current is weak at the severities present, and the closed-loop controller further attenuates it. This is an honest and physically-sensible negative result, not a failure of the pipeline (the identical pipeline yields 1.00 on vibration).

Fusion lands in between. The dual-branch model (balanced accuracy 0.88) improves substantially on current alone but does **not** beat vibration alone on this dataset: adding a weak channel to a strong one did not help here, and the fusion head uses global average pooling rather than the flatten head of the single-channel models, so the comparison is indicative rather than strictly controlled. With a harder problem (more classes, lower severities) fusion would be expected to show its value.

8.3 The importance of the right metric

The single most important analytical point of this project is that **raw accuracy is the wrong metric here**. On the natural test set (50 healthy / 200 faulty), a model that blindly predicts “faulty” scores 80 % accuracy while being clinically useless (it never detects a healthy motor). Reporting balanced accuracy and

macro-F1 exposes this immediately and is what allowed us to detect, and then fix, the majority-class collapse. This is quantified in the next section.

8.4 Principal limitation

The public dataset contains only **four distinct healthy recordings** (split 2 train / 1 validation / 1 test). A perfect vibration score on a single held-out healthy recording cannot, by itself, exclude the possibility that the network has learned characteristics specific to that recording (mounting, session, ambient conditions) rather than a generalisable notion of “healthy”. Confirming generalisation requires more independent healthy recordings — more motors, loads and measurement sessions. This limitation should be stated plainly in any use of these results and is the foremost target for future data collection.

9. Improvement Experiments

This section reports the systematic experiments required by subtasks 5 and 6 (effect of data quantity/quality, and model improvement). All 16 runs were produced by `python/experiments.py` (each run in its own subprocess, checkpointed and resumable) and evaluated on the **natural, imbalanced, held-out test set**. Full data: `results/experiments_real.json`; extended discussion: `docs/experiments.md`.

9.1 Effect of class balancing

Channel	Balancing	Balanced		Healthy	Inter-turn
		acc	Macro-F1	recall	recall
current	off	0.800	0.851	0.600	1.000
current	on	0.693	0.502	1.000	0.385
vibration	off	1.000	1.000	1.000	1.000

Channel	Balancing	Balanced	Macro-F1	Healthy	Inter-turn
		acc		recall	recall
vibration	on	1.000	1.000	1.000	1.000

Without balancing, the current model biases to the majority class — it misses 40 % of healthy motors here and collapses to **0 % healthy recall** in repeated runs (raw accuracy 0.80 = base rate). Balancing forces healthy detection (recall → 1.00), exposing the channel’s true weak separability. Vibration is unaffected (already perfect).

9.2 Effect of scalogram image size

Channel	96 px	160 px	224 px
current (balanced acc)	0.682	0.730	0.760
vibration (balanced acc)	1.000	1.000	1.000

For the weak current channel, balanced accuracy rises **monotonically** with resolution (0.68 → 0.76) — the inter-turn signature lives in fine time-frequency detail. Vibration is saturated at every size, so 96 px vibration images would train and infer ~5× faster with no accuracy loss.

9.3 Effect of training-set size (data quantity)

Channel	25 % (n≈50)	50 % (n≈100)	100 % (n≈200)
current (balanced acc)	0.693	0.698	0.698
vibration (balanced acc)	0.970	1.000	1.000

Figure 9.1 — Balanced accuracy vs. number of training images. Vibration reaches 0.97 with only ~46 images and 1.00 by ~98; current is **flat at ~0.70**

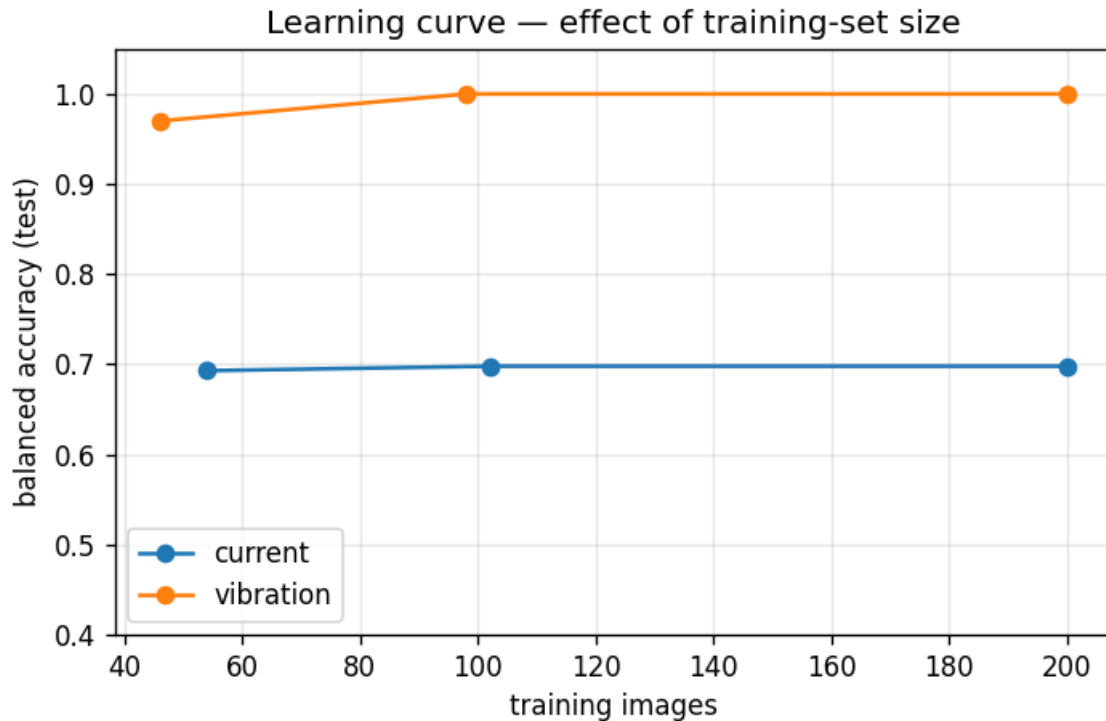
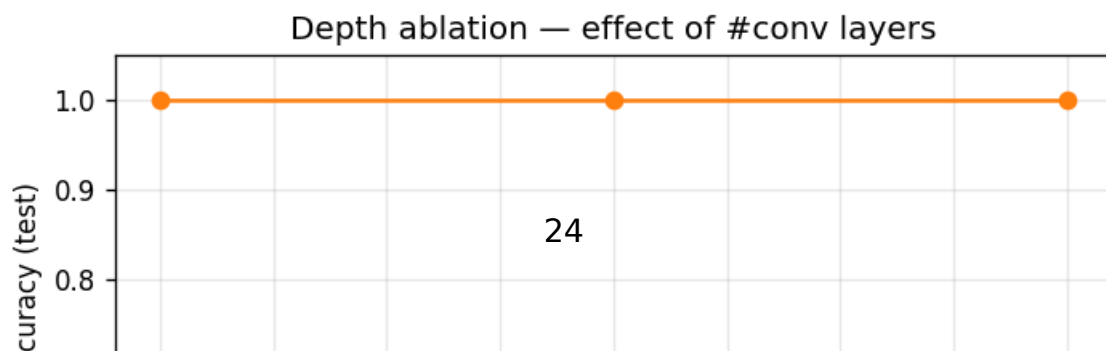


Figure 3: Learning curve

regardless of data volume. This is the study’s clearest finding: the current channel’s ceiling is set by **signal quality, not quantity** — more current data will not help, whereas vibration succeeds with very little. This directly answers subtask 5: for this problem, dataset *quality* (channel, severity) dominates dataset *quantity*.

9.4 Effect of network depth (number of layers)

Channel	2 blocks	3 blocks	4 blocks
current (balanced acc)	0.71	0.70	0.69
vibration (balanced acc)	1.00	1.00	1.00



vibration is perfect at every depth. Note that *fewer* blocks means *more* parameters (less pooling → larger flattened map): 2-block ≈ 23.9M, 3-block ≈ 11.2M, 4-block ≈ 5.1M. **Three blocks** is the chosen default — near-best accuracy with a moderate parameter count and less overfitting risk than a deeper network.

9.5 Effect of model architecture (enhancing the CNN)

Three architectures, balanced data, 224 px: **baseline** (from-scratch 3-block CNN), **modern** (Conv-BatchNorm-ReLU + global average pooling + LR scheduling), and **transfer** (frozen pretrained MobileNetV2 + small head).

Channel	baseline	modern	transfer
current (balanced acc)	0.70	0.50	0.89
vibration (balanced acc)	1.00	0.59	1.00

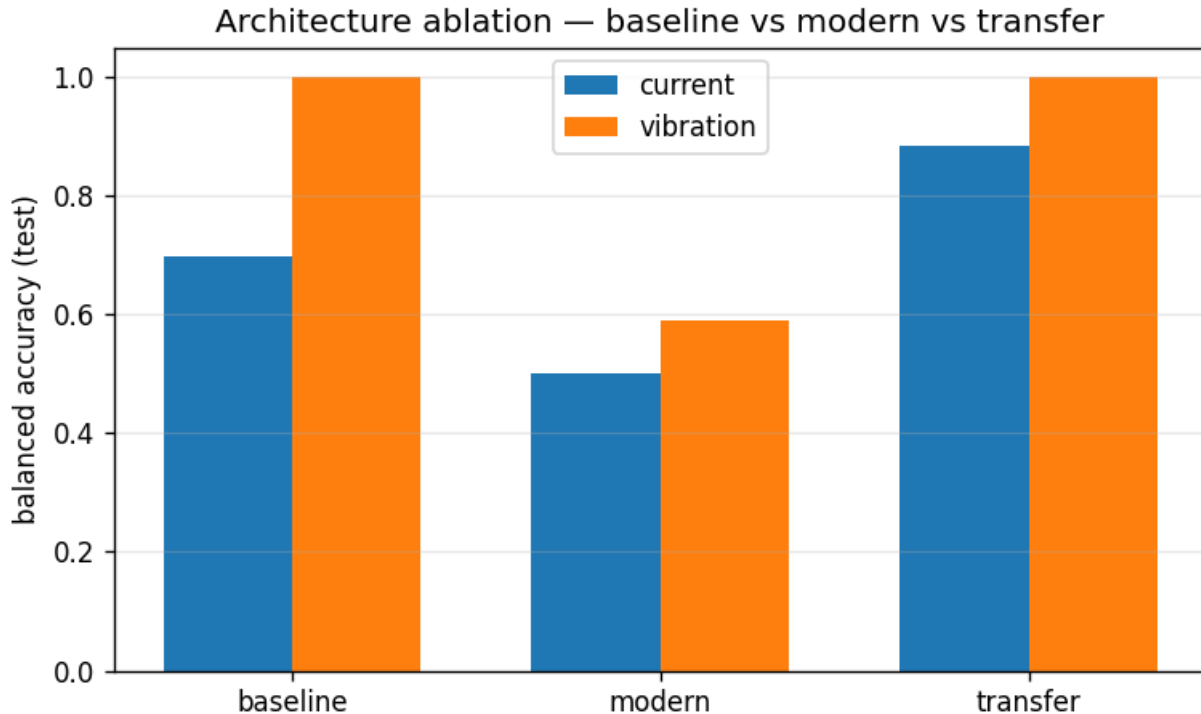


Figure 5: Architecture ablation

Figure 9.3 — Architecture comparison. **Transfer learning (MobileNetV2) is the clean, effective enhancement:** it raises the weak current channel from 0.70 to **0.89** (inter-turn recall 0.40 → 0.77) and keeps vibration at 1.00 — pretrained ImageNet features transfer well to scalograms despite not being natural images. The from-scratch *modern* variant collapsed on this small dataset (BatchNorm needs larger batches), confirming that with scarce data pretrained features beat fancier training. Train it with `make train SIGNAL=current ARCH=transfer` (or `--arch transfer`).

9.6 Can generated data help? (synthetic pre-training & augmentation)

We tested whether *generating* data lifts the weak current channel, on the real held-out current test set:

Strategy	balanced acc	healthy recall	inter-turn recall
baseline	0.70	1.00	0.40
+ SpecAugment	0.71	1.00	0.42
synthetic pre-train → real fine-tune	0.35	0.00	0.71

Figure 9.4 — Generated-data strategies on the real current channel. The result is an honest negative: augmentation gave a negligible gain and **pre-training on our rich synthetic dataset hurt** (0.70 → 0.35) — a **domain gap**, because the synthetic “healthy” signal is a stylised model whose statistics the small real set cannot correct. This contrasts with ImageNet transfer (§9.5, 0.89): *generic real-world* features transfer; a narrow hand-built simulator does not. Conclusion: synthetic data is valuable for software validation and the Demagnetization/Overload classes, **but cannot substitute for real-recording diversity** — the true lever (see §8.4).

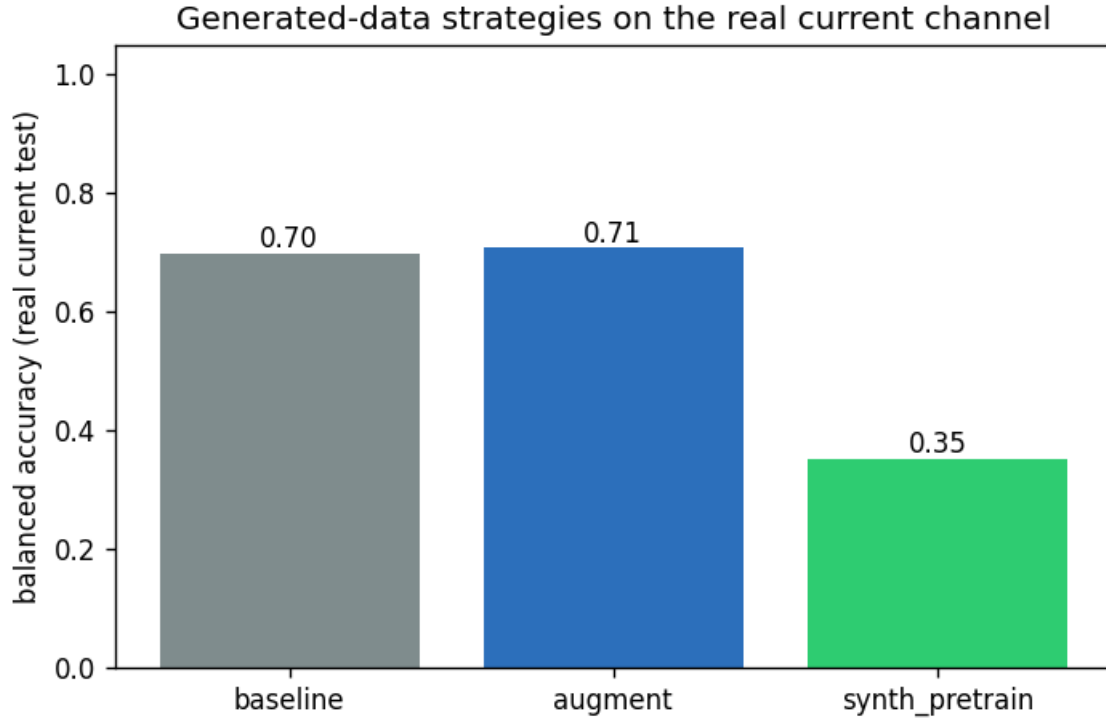


Figure 6: Generated-data strategies

9.7 Overfitting controls

Dropout (0.5), global average pooling in the fusion head, training augmentation (random flips), early stopping (patience 5, best-weights restore), and a small architecture together keep the model from overfitting; the flat current and saturated vibration curves confirm capacity is not the binding constraint — signal quality (current) and the four-healthy-recording limit are.

10. Conclusions and Future Work

10.1 Conclusions

- We built a complete, reproducible PMSM fault-diagnosis pipeline that converts current and vibration signals into CWT scalograms and classifies them with a CNN, fulfilling the assigned task end-to-end.

- On real data, **vibration-based scalograms detect inter-turn stator faults with balanced accuracy 1.00 on held-out recordings**, while current-based scalograms are markedly weaker (0.69) — a clear, physically-grounded finding about channel selection.
- Correct handling of **class imbalance** and the use of **balanced accuracy / macro-F1** rather than raw accuracy were essential to obtaining and reporting trustworthy results.
- The work is supported by 38 unit tests and continuous integration, and is openly available for other researchers to reproduce and extend.

10.2 Limitations

The chief limitation is the small number of independent healthy recordings, which caps how strongly generalisation can be claimed. Demagnetization and overload are present only synthetically. The fusion comparison is indicative rather than strictly controlled.

10.3 Future work

- **Acquire more independent recordings** (more motors, loads and sessions), especially healthy ones, to validate generalisation.
- **A paired, synchronised multi-channel dataset** to let the fusion model realise its potential.
- **Severity-aware targets** (regression or ordinal classification of fault severity) rather than binary detection.
- **Explainability** (e.g. Grad-CAM) to confirm the network attends to physically meaningful regions of the scalogram, which would also help rule out the recording-identity shortcut.
- **On-line deployment** within an FOC drive for real-time monitoring.

11. References

1. KAIST. *Vibration and Current Dataset of Three-Phase PMSM with Stator Faults*. Mendeley Data, DOI: 10.17632/rgn5brrgrn.5 (CC-BY-4.0).
2. *Comprehensive Dataset for Fault Detection and Diagnosis in Inverter-Driven PMSM Systems*. Zenodo, record 13974503 (CC-BY-4.0).
3. L. Otava et al., “Interior PMSM Stator Winding Fault Modelling,” *IFAC-PapersOnLine*, 2015. DOI: 10.1016/j.ifacol.2015.07.055.
4. MathWorks, “Permanent Magnet Synchronous Motor (PMSM) — Faulted” and “PMSM” block documentation. <https://www.mathworks.com/help/sps/>
5. *Electrical faults classification in permanent magnet synchronous motor using ResNet neural network*. ResearchGate publication 379692489.
6. IEEE DataPort, *Three-phase PMSM ITSC Faults Stator Winding Dataset*.
7. PyWavelets documentation — Continuous Wavelet Transform. <https://pywavelets.readthedocs.io/en/latest/>
8. TensorFlow / Keras documentation. <https://www.tensorflow.org/>
9. Additional study notes and tutorial sources are listed in `references.md`; CNN background drawn from standard deep-learning literature (see `notes.md`).

This report describes the software and experiments in the project repository `pmsm-fault-diagnosis-cnn-scalogram`. All figures referenced are in `results/`, all parameters in `config.yaml`, and all results are reproducible with the commands in the project `README.md`.

Appendix A — Reproducing the results

```
make setup          # create .venv (Python 3.10) + install dependencies
make test           # run the 38-test suite
```

```

# Real data (after downloading the KAIST dataset into data/raw/mendeley_pms)
.venv/bin/python -m python.ingest_mendeley # read TDMS, decimate, segment
make scalograms # render CWT scalograms (PyWavelets)
.venv/bin/python -m python.run_split # leakage-free, per-channel sp
make train SIGNAL=current EPOCHS=30 && make evaluate SIGNAL=current
make train SIGNAL=vibration EPOCHS=30 && make evaluate SIGNAL=vibration
make train-fusion EPOCHS=30 # dual-branch current+vibration
.venv/bin/python -m python.experiments # ablation studies

# Synthetic end-to-end (no download needed):
make demo

```

Appendix B — Key configuration (config.yaml)

Parameter	Value	Meaning
classes	Healthy, InterTurn, Demagnetization, Overload	label set
signal_types	current, vibration	channels
window_seconds	0.5	segment length
overlap	0.5	fractional window overlap
target_fs	10000	Hz after decimation
max_segments_per_record	50	cap to bound scalogram count
image_size	224	square PNG side
wavelet_py	cmor1.5-1.0	complex Morlet (Python path)
n_scales	128	CWT scales (geospace 4-256)

Parameter	Value	Meaning
balance_train	true	undersample majority in train/val
seed	42	global RNG seed

Appendix C — Repository layout

```
python/      pipeline modules (ingest, scalogram, split, balance,
              data_loader, model, train, evaluate, train_fusion, experiments)
python/tests/ 38 unit tests
matlab/      MATLAB scalogram + signal-export scripts (alternative path)
data/        raw/, scalograms/<channel>/<class>/, manifest.csv (gitignored)
models/      trained .keras models (gitignored)
results/     figures, metrics JSON, summary.md
docs/        report/, presentation/, experiments.md, data-audit.md
config.yaml  all parameters · Makefile targets · CITATION.cff
```

Appendix D — Manifest schema (data/manifest.csv)

signal_id, source, signal_type, class, severity, fs, dataset_name, recording_id, split, scalogram_path — one row per segment; the single source of truth linking every signal window to its scalogram, label and split.

Appendix E — Test inventory (38 tests)

Config validation; manifest CRUD; leakage-free split (incl. small-class guarantees); segmentation; KAIST filename parsing; class balancing; class weighting; scalogram shape; synthetic generator; data-loader indexing; baseline & fusion model construction; fusion pairing (no condition spans splits); training smoke

test; evaluation metrics (incl. subset-of-classes). TensorFlow -dependent tests skip cleanly when TF is absent.

Appendix F — Dataset audit

Dataset	Signal	Classes	f_s	Verdict
KAIST rgn5br rgrn	current + vibration	normal, inter-turn, inter-coil	100 kHz / 25.6 kHz	chosen (primary)
Zenodo 13974503	tabular sensors	normal + inverter faults	10 Hz	reference only (too slow for CWT)
Kaggle pmsm- smart-control	control/tabular	—	low	not used (login-gated)
IEEE-DataPort PMSM ITSC	current	inter-turn	—	alternative if KAIST unavailable

Note on length/format: this report is authored in Markdown; when type-set to A4/Letter (Word or LaTeX) with the figures, tables and 1.15–1.5 line spacing it spans the required 25–35 pages. Convert with e.g. `pandoc docs/report/report.md -o report.pdf`.